Statement of Purpose for Doctor of Philosophy in Biomedical Engineering

With profound enthusiasm and unwavering commitment to the interdisciplinary field of biomedical engineering, I am delighted to submit my application for the Doctor of Philosophy (PhD) program in Biomedical Engineering at the National University of Singapore (NUS). Over a decade of hands-on experience as a software algorithm engineer and interdisciplinary academic exploration has reinforced my conviction that the deep integration of engineering technology and biomedicine holds the key to addressing complex healthcare challenges. As a world-renowned institution at the forefront of this field, NUS, with its exceptional academic resources, cutting-edge research directions, and open interdisciplinary environment, represents the ideal platform to realize my academic aspirations and professional transformation.

I. Professional Accomplishments: Practical Exploration of Engineering Technology and Interdisciplinary Applications

Over the past ten years, I have dedicated myself to the field of software algorithm development, specializing in computer vision, artificial intelligence (AI), and system integration, accumulating comprehensive experience spanning from technology R&D to project implementation. As a core technical expert and project leader, I have spearheaded numerous industry-influential projects: In the vision-guided robotic arm project, I integrated a Transformer model to enable action prediction and real-time environmental adaptation, constructing a seamless interaction framework between visual, auditory, and motor systems. The high-precision optical character recognition (OCR) tool I developed for the semiconductor industry achieved performance on par with or exceeding that of industry leader Cognex, boasting a 100% recognition rate in fixed-format scenarios. Within the biomedical realm, I participated in designing reagent bottle seal inspection systems and medical character recognition solutions, successfully applying AI algorithms to production quality control and medical data processing—experiences that vividly demonstrated the immense potential of engineering technology to empower biomedical advancements. These practices not only honed my proficiency in C/C++/Python programming, AI model deployment (PyTorch, ONNX Runtime), and system optimization but also deepened my understanding of how efficient engineering solutions can provide robust support for biomedical research and clinical applications. This realization has become the core driving force behind my decision to transition into the field of biomedical engineering. Additionally, I have filed for or obtained more than 10 invention patents throughout my career, forming a complete thought cycle from technological innovation to achievement transformation.

II. Academic Advancement: Construction and Enhancement of an Interdisciplinary Knowledge System

To systematically enter the field of biomedical engineering, I proactively expanded my interdisciplinary perspective during my Master's studies in Microelectronics Technology & Materials at The Hong Kong Polytechnic University. Under the guidance of my supervisor, I delved into molecular dynamics simulation techniques, proficiently mastering specialized tools such as LAMMPS and AI platforms, and completed research on the phase transition mechanism of perovskite materials—laying a solid theoretical and experimental foundation for interdisciplinary research. Recognizing the gaps in my knowledge of core foundational subjects such as cell biology and molecular biology, I have initiated a structured self-learning plan: I am currently studying Professor Rongwu Yang's Molecular Biology course and plan to proceed with Professor Zhonghe Zhai's Cell Biology thereafter to bridge these gaps and fully prepare for integration into biomedical engineering research. My undergraduate education in Microelectronics at Xi'an University of Technology, coupled with awards in provincial and university-level academic competitions, has further equipped me with a robust engineering foundation and rigorous scientific thinking.

III. Research Vision: Exploring Directions for Engineering Technology to Empower Biomedicine

Building on my technical expertise and academic planning, I aim to contribute to the following research areas during my PhD studies at NUS:

1. Optimization of Medical and Biological Image Processing: Leveraging over a decade of experience in image processing, I seek to deeply integrate AI algorithms with biological imaging data to develop automated analysis tools. These tools will encompass target segmentation and extraction, localization and counting, defect detection and classification, as well as dynamic target tracking—ultimately enhancing the efficiency and accuracy of biomedical experiments.
2. Construction of Hybrid Computing Platforms: Drawing on the parallel architecture of "traditional calibration + AI prediction" from the robotic arm project, I intend to develop an adaptive computing platform tailored for biological simulations. This platform will integrate real-time data collection to improve AI model accuracy while comparing the performance of traditional methods and AI solutions to select stable and reliable workflows for specific tasks, thereby supporting cutting-edge research such as bioengineering-based digital twins.
3. Optimization of Computational Tools and Hardware-Software Integration: Utilizing my programming and system tuning capabilities, I aim to participate in the optimization and upgrading of commonly used molecular dynamics tools in laboratories (e.g., GROMACS). Specific optimizations will include modifying source code to enhance tool adaptability to specific research scenarios, such as implementing CUDA-based GPU acceleration and adding preprocessing/postprocessing logic. Additionally, leveraging my experience in PCB design, 3D printing, and wireless communication technology, I hope to assist in the development of miniaturized medical devices and data transmission

systems, facilitating the system-level integration of research in areas such as microfluidic organoid culture and diagnostic systems.

IV. Alignment with NUS: A Mutual Pursuit with the Department of Biomedical Engineering

The Department of Biomedical Engineering at NUS, with its profound legacy in interdisciplinary research and dual emphasis on technological innovation and clinical translation, aligns perfectly with my research vision. I am confident that my technical strengths in AI algorithms, system development, and interdisciplinary project management—combined with my proactive learning attitude and multilingual communication skills (proficiency in Cantonese, Mandarin, and English, with basic Japanese)—will bring a unique engineering perspective to the department's research teams, fostering technological innovation and the translation of research outcomes in biomedicine. Meanwhile, I eagerly anticipate deepening my biomedical expertise under the guidance of NUS's world-class faculty, transitioning from an engineering specialist to an interdisciplinary biomedical engineering researcher, and ultimately growing into a versatile professional capable of driving technological breakthroughs and clinical applications in this field.

I am fully prepared to dedicate myself wholeheartedly to PhD research and study. I aspire to thrive in NUS's academic environment, delve deeply into biomedical engineering, and contribute to improving human health and advancing medical technology through the innovative integration of engineering and biomedicine. I sincerely look forward to the opportunity to join the Department of Biomedical Engineering at NUS and explore the boundless possibilities of interdisciplinary research alongside leading scholars and peers.